

ST BEES HEAD, United Kingdom

WMO: 032100

Lat: 54.5178N

Lon: 3.6147W

Elev: 404

StdP: 14.48

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	28.1	30.2	21.2	16.2	32.7	23.6	18.1	33.0	43.5	40.4	40.1	40.7	8.2	30	0.766

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.4	71.1	62.6	67.6	61.1	64.8	59.7	64.0	68.8	62.3	66.0	60.9	63.7	10.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.2	84.9	65.4	60.8	80.7	63.5	59.6	77.3	62.2	29.4	69.1	28.2	66.2	27.2	63.8	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
36.7	32.2	28.8	DB	25.6	77.1	2.7	3.9	23.6	79.9	22.1	82.2	20.6	84.4	18.6	87.2
			WB	24.4	67.8	2.3	1.9	22.7	69.2	21.4	70.3	20.1	71.4	18.4	72.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.7	40.0	40.0	42.0	45.8	50.8	55.1	58.5	58.7	55.6	50.9	45.2	41.4
	DBStd	8.03	4.40	4.26	4.08	4.39	4.71	3.80	3.60	3.15	3.28	4.18	4.69	5.18
	HDD50	1485	309	280	250	140	42	3	0	0	2	40	151	268
	HDD65	5958	774	700	714	577	441	298	208	201	282	438	593	732
	CDD50	1013	0	0	1	13	67	156	263	268	171	66	8	0
	CDD65	12	0	0	0	0	1	1	6	4	0	0	0	0
	CDH74	43	0	0	0	0	5	3	21	14	0	0	0	0
	CDH80	3	0	0	0	0	0	0	2	1	0	0	0	0
	WSAvg	14.2	16.7	16.0	14.3	13.1	12.5	12.0	11.5	12.6	13.9	15.9	16.3	16.0
	PrecAvg	45.9	4.4	3.2	3.4	2.6	2.8	2.9	3.2	3.9	4.1	5.4	4.9	4.9
(15) Precipitation	PrecMax	57.2	6.9	6.3	5.6	4.5	4.9	7.6	5.7	8.4	7.0	9.8	9.8	11.3
	PrecMin	35.7	0.5	0.8	1.7	0.7	0.5	1.3	1.9	0.7	0.9	1.5	2.8	1.4
	PrecStd	6.7	1.6	1.6	1.1	1.0	1.1	1.3	1.0	1.9	1.4	2.7	1.8	2.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.0	51.2	56.2	64.7	71.8	73.3	76.8	75.6	68.3	62.5	56.5	52.1
		MCWB	49.0	46.4	49.2	54.0	61.3	63.3	65.2	65.4	61.6	57.6	54.4	51.0
	2%	DB	48.2	48.2	51.5	58.9	66.3	67.6	71.3	69.1	64.4	59.4	54.5	50.7
		MCWB	47.2	46.4	46.9	51.0	58.9	60.7	63.3	62.4	59.3	56.9	53.1	49.8
	5%	DB	47.1	46.8	48.6	54.5	61.6	63.9	67.3	65.7	62.1	57.8	52.9	49.4
		MCWB	46.1	45.4	46.1	49.2	55.4	58.7	61.6	60.8	58.4	55.9	51.7	48.4
	10%	DB	45.9	45.6	47.0	51.7	58.1	61.2	64.3	63.3	60.5	56.6	51.7	48.1
		MCWB	44.9	44.4	45.1	47.9	53.6	57.1	59.9	59.5	57.5	54.7	50.5	47.0
	0.4%	WB	49.3	48.4	50.6	55.2	62.6	64.5	67.0	66.3	63.5	59.7	55.4	51.5
		MCDB	49.5	49.2	53.4	63.5	68.6	69.6	74.0	72.1	66.3	60.8	55.6	51.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.6	46.9	48.1	52.0	59.3	61.5	64.3	63.5	61.0	57.8	53.7	50.3
		MCDB	48.0	47.6	50.0	56.6	66.0	66.3	69.6	67.6	62.9	58.6	54.2	50.6
	5%	WB	46.5	45.8	46.7	50.1	56.5	59.5	62.4	61.9	59.6	56.5	52.0	48.8
		MCDB	46.9	46.5	48.2	53.4	60.6	63.3	66.5	64.6	61.2	57.4	52.7	49.3
	10%	WB	45.2	44.7	45.6	48.6	54.2	57.6	60.6	60.5	58.3	55.2	50.7	47.3
		MCDB	45.8	45.4	46.9	51.1	57.3	60.5	63.5	62.6	59.9	56.3	51.5	48.0
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	6.4	7.7	9.2	10.1	9.3	8.8	8.4	8.3	7.4	6.4	6.4
		MCDBR	6.0	6.2	8.7	13.2	15.7	13.6	13.5	11.9	10.2	8.0	6.8	6.6
		MCWBR	6.6	5.8	6.9	8.9	10.7	9.1	9.1	7.9	7.5	7.2	7.4	7.1
	5% WB	MCDBR	6.1	5.6	7.6	10.8	13.8	12.3	12.0	9.8	8.7	7.1	6.6	6.5
		MCWBR	6.7	5.7	6.6	7.9	9.9	8.9	8.4	7.3	7.6	7.2	7.5	7.3
(40) Clear-Sky Solar Irradiance	taub	0.325	0.335	0.360	0.378	0.382	0.378	0.379	0.367	0.356	0.342	0.319	0.310	
	taud	2.288	2.376	2.343	2.321	2.361	2.396	2.394	2.443	2.475	2.457	2.365	2.264	
	Ebn at Noon	193	238	257	269	274	276	273	270	258	233	196	170	
	Edh at Noon	19	24	30	36	36	35	35	31	27	22	17	16	
(44) All-Sky Solar Radiation	RadAvg	175	389	747	1193	1547	1562	1508	1217	860	460	224	134	
	RadStd	23	43	68	114	120	154	153	89	73	55	23	13	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (62 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page